

Yiqi (Verna) Tian

yit30@pitt.edu

Homepage

Google Scholar

Last Update: 09/11/2026

Research Interests

Methodologies: Data-driven decision-making; Robust optimization, Distributionally robust optimization.

Applications: Healthcare operations management; Diffusion models; Trustworthy AI.

Education

University of Pittsburgh, Pittsburgh, USA

August 2020 – July 2026

Ph.D. in Operations Research, Department of Industrial Engineering

GPA: 4.00 / 4.00

Dissertation: *Distributionally Robust Optimization: Theory, Algorithms, and Applications in Healthcare and Trustworthy AI*

Advisor: Prof. Bo Zeng

University of Pittsburgh, Pittsburgh, USA

August 2018 – April 2020

B.S. in Industrial Engineering, Minor in Mathematics

GPA: 3.96 / 4.00

Awards: Summa Cum Laude; Honor Student

Sichuan University – Pittsburgh Institute (SCUPI), Chengdu, China

September 2016 – July 2018

B.S. in Industrial Engineering

GPA: 3.93 / 4.00

Awards: Outstanding Sichuan Province College Graduates; Outstanding Sichuan University Graduates

Research Experience

Harvard Medical School, Boston, USA

July 2024 – Present

Graduate Research Assistant at CAMCA, Massachusetts General Hospital

BLOODSAFE program, Pittsburgh, USA & Nairobi, Kenya

June 2021 – June 2026

Graduate Research Assistant for Modeling and Simulation

Honors & Awards

- **First Place**, “Doing Good with Good O.R.” Paper Competition, INFORMS (2024)
- **Teaching Assistant of the Year**, Department of Industrial Engineering, UPitt (2024)
- **Best Paper Award**, International Joint Conference on IE and OM (2023)
- **Research Assistant of the Year**, Department of Industrial Engineering, UPitt (2023)
- **Outstanding Sichuan Province College Graduate**, 1% of 220k grads in Sichuan Province (2020)
- **Outstanding Sichuan University Graduate**, one of two winners out of 160 SCUPI graduates (2020)
- **Allias/Holzman Teaching Fellowship**, Department of Industrial Engineering, UPitt (2020)
- **Undergraduate Summer Research Fellowship**, Swanson School of Engineering, UPitt (2019)
- **SCUPI Excellent Peer Advisor** (2018)
- **China National Scholarship**, 0.2% among 7 million college students (2017, 2018)
- **Sichuan University Excellent Student** (2017, 2018)
- **SCUPI Queen of the Year** (2017)

Publications

Journal Paper

- J1. “Strengthening Blood Transfusion System in Kenya by Analytic Tools”. **Yiqi Tian**, Bo Zeng, Jayant Rajgopal, Bopaya Bidanda, Pratap Kumar, Juan Carlos Puyana.
 - **First Place** of the 2024 INFORMS “[Doing Good with Good O.R.](#)” Paper Competition
 - Presented at 2024 INFORMS annual conference, 2025 RIT Graduate Seminar Series
 - Minor revision at *INFORMS Journal on Applied Analytics*
- J2. “A Primal Perspective on Distributionally Robust Optimization: An Investigation on Modeling and Solution Strategies”. **Yiqi Tian**, Bo Zeng.
 - Under review
- J3. “Simulating the blood transfusion system in Kenya: Modelling methods and exploratory analyses.” **Yiqi Tian**, Bo Zeng, Jana MacLeod, *et al.*
PLOS Global Public Health 5.8 (2025): e0004587
 - Presented at the 2023 Consortium of Universities for Global Health Conference, Washington, D.C.
 - Presented at the BLOODSAFE 2025 Symposium, Nairobi, Kenya
 - Social Media Post: [Pitt Engineering News](#)
- J4. “Protocol: identifying policy, system, and environment change interventions to enhance availability of blood for transfusion in Kenya, a mixed-methods study.” Alejandro Munoz-Valencia, Jackline O. Aridi, Linda S. Barnes, **Yiqi Tian**, *et al.*
BMC Health Services Research, 23:963 (2023).
- J5. “The BLOODSAFE program: Building the future of access to safe blood in Sub-Saharan Africa.” Meghan Delaney, Susan Telke, Shimian Zou, **Yiqi Tian**, *et al.*
Transfusion, 62(11):2282–2290 (2022).

Conference Paper

- C1. “RODS: Robust Optimization Inspired Diffusion Sampling for Detecting and Reducing Hallucination in Generative Models”. **Yiqi Tian**[†], Pengfei Jin[†], Mingze Yuan, Na Li, Bo Zeng, Quanzheng Li.
NeurIPS 2025 [Code] [Poster, 24.5% acceptance rate]

Work in Progress

1. “Hallucination-Aware Diffusion Sampling for Inverse Problems via Robust Prior Updates”. Pengfei Jin[†], **Yiqi Tian**[†], Kailong Fan, Bingjie Qi, Quanzheng Li.
2. “DuetFair: Coupling Inter- and Intra-Subgroup Robustness for Fair Medical Image Segmentation”. **Yiqi Tian**[†], Sangjoon Park[†], Bo Zeng, Pengfei Jin, Yujin Oh, Quanzheng Li.
3. “Enhancing Blood Transfusion System In Africa: Motivating Donations with Two-stage Utility-Based Blood Drive Planning and Strategic Advertising”. **Yiqi Tian**, Bo Zeng, Jayant Rajgopal, Bopaya Bidanda, Pratap Kumar, Juan Carlos Puyana.
 - Presented at 2023, 2024 INFORMS annual conference, [2025 Purdue SCOM conference](#)

[†] Equal contribution.

– Prepared for submission to *Production & Operations Management*

4. “Optimizing Radiologist Staffing with Uncertainty-Aware, Data-Driven Scheduling for On-Time Scan Reads”. **Yiqi Tian**, Melissa Viator, Quanzheng Li, Bo Zeng, Oleg Panykh

Other Projects

1. “Parameter-Refined PBPK Modeling for Enhanced Dosimetry in ^{177}Lu -PSMA Radiopharmaceutical Therapy Using Estimated and Personalized Parameters” – **Accepted as Oral Presentation** in 2025 *Society of Nuclear Medicine and Molecular Imaging*.
Yiqi Tian, Sifan Song, Robert Seifert, Christoph Clement, Quanzheng Li, Axel Rominger, Kuangyu Shi, Bo Zeng, Ning Guo.
2. “Using Industrial Engineering to Improve Blood Collection and Delivery in a Low/Middle Income Country: A Case Study in Africa”. – **Best Paper Award** at 2023 *International Joint Conference on Industrial Engineering and Operations Management*
Yiqi Tian, Bopaya Bidanda, Jayant Rajgopal, Bo Zeng.

Teaching Experience

Instructor

- ENGR 0021 – Probability & Statistics for Engineers, Spring 2024: 95% response (39/41), overall 4.33/5
- ENGR 0021 – Probability & Statistics for Engineers, Spring 2023: 69% response (25/36), overall 3.72/5

Teaching Assistant

- IE 1080 – Supply Chain Analysis (*Undergrad*), Fall 2023
- IE 2100 – Supply Chain Analysis (*Grad*), Fall 2022; Fall 2020
- IE 2001 – Operations Research (*Grad*), Spring 2022
- IE 1055/2055 – Facility Layout & Material Handling (*Undergrad / Grad*), Fall 2021
- ENGR 0021 – Probability & Statistics (*Undergrad*), Spring 2021; Summer 2023
- IE 1071 – Statistical Testing & Regression (*Undergrad*), Spring 2020

Skills

OR Tools	Gurobi, CPLEX, SIMIO, Visio
AI Tools	PyTorch, TensorFlow, OpenAI Gym
Biomedical Modeling & Analysis	PK-Sim, MoBi, 3D Slicer
Programming Languages	Python, Julia, R, MATLAB
Languages	Mandarin (native), English (fluent)

Reference

Dr. Bo Zeng

Associate Professor
Department of Industrial Engineering, University of Pittsburgh
Email: bzeng@pitt.edu

Dr. Quanzheng Li

Director, Center for Advanced Medical Computing and Analysis (CAMCA)
Associate Professor, Dept. of Radiology, Harvard Medical School & Massachusetts General Hospital
Email: li.quanzheng@mgh.harvard.edu

Dr. Juan C. Puyana

Chair of Global Surgery, Royal College of Surgeons in Ireland, Institute of Global Surgery, School of Population Health
Principal Investigator, Pathways for Innovation in Blood Transfusion Systems in Kenya, University of Pittsburgh
Email: puyajc@UPMC.EDU